University of Jeddah
College of Computer Science and Engineering
Department of Information Systems & Technology

Impact of Study Hours on GPA

Linear Regression Analysis

CCDS211-Introduction to Data Science

Student Name:

Dalia Mossa Alshamrani 2311355

Introduction

This study explores the relationship between students' weekly study time and their academic performance, as measured by Grade Point Average (GPA). Academic success is a multifaceted outcome influenced by various factors, including individual study habits, cognitive abilities, motivation, and socioeconomic background. Among these, the amount of time a student dedicates to studying outside of class is often considered an essential factor contributing to academic achievement. Understanding how study time correlates with GPA can provide valuable insights for educators, students, and policymakers, helping to guide recommendations for effective study habits so **GPA : the dependent variable , StudyTimeWeekly : the independent variable.**

In this analysis, a dataset of 2,392 students, **downloaded from** Kaggle, was examined to determine the extent to which weekly study hours predict academic performance. By visualizing the data, computing summary statistics, and performing correlation and regression analyses, this study seeks to quantify the relationship between study time and GPA. While prior research generally suggests a positive association between study time and academic outcomes, the strength and nature of this relationship remain an open question, particularly given the influence of other factors. This study aims to clarify whether a significant relationship exists and, if so, how impactful weekly study time is on students' GPAs. The findings may contribute to a more nuanced understanding of study habits in academic success.

Results

No data cleaning was required in this analysis because the dataset was already in a well-prepared and structured format suitable for statistical modeling. First, a check for missing data revealed no missing values. This meant there was no need to impute or remove incomplete observations, ensuring that the dataset could be used as-is for analysis. Additionally, the selected variables, **StudyTimeWeekly** and **GPA**, were directly relevant to the research question and were continuous, making them appropriate for the linear regression model. The values of these variables were within plausible ranges: **StudyTimeWeekly** ranged from ~0 to ~20 hours, which is consistent with realistic study habits, and **GPA** ranged from 0 to 4, adhering to the standard grading system. The boxplots on Figure 3 confirmed the absence of extreme outliers that could distort the analysis, further validating the quality of the data. Furthermore, no duplicate rows were identified, as the dataset consisted of individual records for student performance, inherently precluding redundancy. The dataset's structure required no renaming of columns, conversion of data types, or resolution of inconsistencies, which ensured that it was ready for immediate analysis without additional preprocessing steps.

A scatter plot was created to visualize the relationship between study time and GPA and presented on FIgure 1. The plot, supplemented by a smoothing line, suggests a positive trend, indicating that as weekly study time increases, GPA tends to improve, albeit with considerable variation. A density plot, Figure 2, for each variable highlights their distributions. Study time

exhibits a slightly negatively skewed distribution, while GPA appears moderately negatively skewed as well, with a peak around a median GPA of 1.893. This distribution suggests that a significant portion of students achieve GPAs near the median, with fewer students at the extremes. Descriptive statistics reveal that weekly study time ranges from approximately 0 to 20 hours, with a mean of 9.77 hours. Similarly, GPA ranges from 0 to 4.0, with a mean of 1.91. The median values for study time and GPA are 9.71 hours and 1.89, respectively, indicating relatively symmetrical distributions.

Codes-1 (1).R* × dat ×

Filter

	StudentID	Age	Gender	Ethnicity	ParentalEducation	StudyTimeWeekly	Absences	Tutoring	ParentalSupport	Extracurricular	Sports	Music	Volun
1	1001	17	1	0	2	19.833722808	7	1	2	0	0	1	
2	1002	18	0	0	1	15.408756056	0	0	1	0	0	0	
3	1003	15	0	2	3	4.210569769	26	0	2	0	0	0	
4	1004	17	1	0	3	10.028829474	14	0	3	1	0	0	
5	1005	17	1	0	2	4.672495273	17	1	3	0	0	0	
6	1006	18	0	0	1	8.191218545	0	0	1	1	0	0	
7	1007	15	0	1	1	15.601680475	10	0	3	0	1	0	
8	1008	15	1	1	4	15.424496306	22	1	1	1	0	0	
9	1009	17	0	0	0	4.562007558	1	0	2	0	1	0	
10	1010	16	1	0	1	18.444466363	0	0	3	1	0	0	
11	1011	17	0	0	1	11.851363655	11	0	1	0	0	0	
12	1012	17	0	0	1	7.598485819	15	0	2	0	0	0	
13	1013	17	0	1	1	10.038711616	21	0	3	1	0	0	
14	1014	17	0	1	2	12.101425069	21	0	4	0	1	0	
15	1015	18	1	0	1	11.197810637	9	1	2	0	0	0	
16	1016	15	0	0	2	9.728100711	17	1	0	0	1	0	
17	1017	18	0	3	1	10.098656082	14	0	2	1	1	0	
18	1018	18	1	0	0	3.528238209	16	1	2	0	0	0	
19	1019	18	0	1	3	16.254658086	29	0	2	1	0	0	
20	1020	17	0	0	1	10.835206399	9	0	2	0	0	1	
21	1021	16	1	0	3	2.621597234	2	0	3	0	0	0	
22	1022	15	0	0	2	15.323142032	25	0	1	1	0	0	
23	1023	16	1	1	0	18.648879568	29	1	1	0	0	0	
24	1024	18	1	3	4	18.946137985	20	0	2	1	0	0	
25	1025	18	1	0	1	7.380354648	15	0	2	0	0	0	

Summary of Dataset:

```
> summary(dat)
   StudentID         Age            Gender         Ethnicity      ParentalEducation StudyTimeWeekly        Absences
 Min.   :1001   Min.   :15.00   Min.   :0.0000   Min.   :0.0000   Min.   :0.000     Min.   : 0.001057   Min.   : 0.00
 1st Qu.:1599   1st Qu.:15.00   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.000     1st Qu.: 5.043079   1st Qu.: 7.00
 Median :2196   Median :16.00   Median :1.0000   Median :0.0000   Median :2.000     Median : 9.705363   Median :15.00
 Mean   :2196   Mean   :16.47   Mean   :0.5109   Mean   :0.8775   Mean   :1.746     Mean   : 9.771992   Mean   :14.54
 3rd Qu.:2794   3rd Qu.:17.00   3rd Qu.:1.0000   3rd Qu.:2.0000   3rd Qu.:2.000     3rd Qu.:14.408410   3rd Qu.:22.00
 Max.   :3392   Max.   :18.00   Max.   :1.0000   Max.   :3.0000   Max.   :4.000     Max.   :19.978094   Max.   :29.00
    Tutoring       ParentalSupport Extracurricular      Sports           Music          Volunteering         GPA
 Min.   :0.0000   Min.   :0.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.000
 1st Qu.:0.0000   1st Qu.:1.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.175
 Median :0.0000   Median :2.000   Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000   Median :1.893
 Mean   :0.3014   Mean   :2.122   Mean   :0.3834   Mean   :0.3035   Mean   :0.1969   Mean   :0.1572   Mean   :1.906
 3rd Qu.:1.0000   3rd Qu.:3.000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:2.622
 Max.   :1.0000   Max.   :4.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :4.000
   GradeClass
 Min.   :0.000
 1st Qu.:2.000
 Median :4.000
 Mean   :2.984
 3rd Qu.:4.000
 Max.   :4.000
```

```
> # Load packages
> library(tidyverse)
>
> # Import data
> dat <- read.csv("C:\\Users\\Lenovo\\Desktop\\Student_performance_data _.csv")
> # Select relevant columns
> df <- dat %>% select(StudyTimeWeekly, GPA)
> str(df)
'data.frame':   2392 obs. of  2 variables:
 $ StudyTimeWeekly: num  19.83 15.41 4.21 10.03 4.67 ...
 $ GPA            : num  2.929 3.043 0.113 2.054 1.288 ...
>
> # Check na values
> sum(is.na(df))
[1] 0
```

```
'data.frame':   2392 obs. of  2 variables:
 $ StudyTimeWeekly: num  19.83 15.41 4.21 10.03 4.67 ...
 $ GPA            : num  2.929 3.043 0.113 2.054 1.288 ...

[1] 0
```

The analysis of student performance data includes 2,392 observations, covering several important variables. Weekly study hours (StudyTimeWeekly) range from 0.001 to 19.97 hours, with a mean of 9.77, a first quartile of 5.04 hours, and a median of 9.70 hours, indicating that half of the students study less than 10 hours weekly. The GPA ranges from 0 to 4, with a mean of 1.91 and a median of 1.89, reflecting variability in academic performance, where 25% of students score below 1.17. Absences (Absences) reach a maximum of 29 days, with a mean of 14.54 days and a median of 15 days, showing high absenteeism for some students, which may affect their commitment to studies. Parental education levels (ParentalEducation) range from 0 (low) to 4 (high), with a mean of 1.75, indicating a moderate level of parental academic support. Student ages range from 15 to 18 years, with a mean of 16.47, and gender is almost evenly distributed (mean: 0.51). Participation in extracurricular activities like sports, music, and volunteering is low to moderate, with rates between 19% and 38%. Finally, the data has no missing values, making it clean and ready for in-depth analysis to identify factors influencing student performance.

```
> # Scatterplot of variables
> stm <- df$StudyTimeWeekly ; gpa = df$GPA
>
> scatter.smooth(stm, gpa, main = "Scores ~ Study time", lpars =
+                    list(col = "red", lwd = 3, lty = 3),
+                xlab = "Study Time (Weekly)", ylab = "GPA")
```

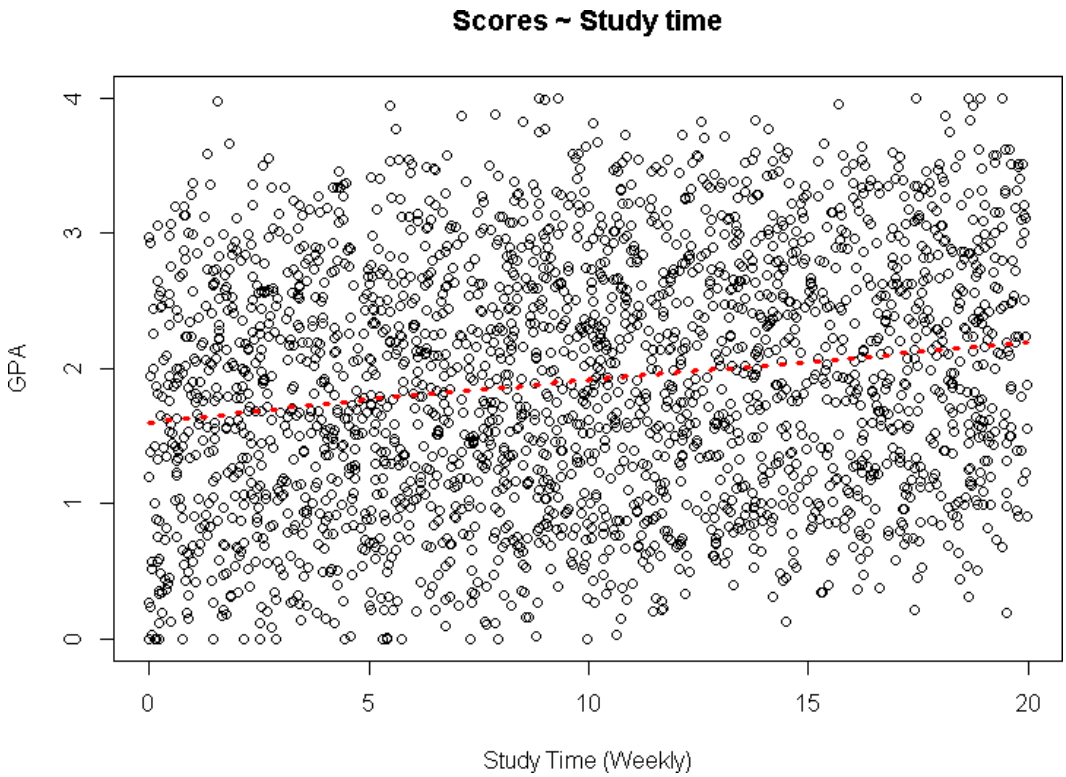

Figure 1: Scatterplot of Study time and GPA

The scatterplot shows a weak positive relationship between weekly study time and GPA. The red trendline indicates that as study hours increase, GPA slightly improves, but the large spread of data points around the trendline suggests other significant factors influencing GPA.

```
> # Density plots
> par(mfrow = c(1, 2))
>
> plot(density(df$StudyTimeWeekly), main = "Study Time", ylab = "Frequency")
> polygon(density(df$StudyTimeWeekly), col = "green")
> plot(density(df$GPA), main = "GPA", ylab = "Frequency")
> polygon(density(df$GPA), col = "red")
>
```

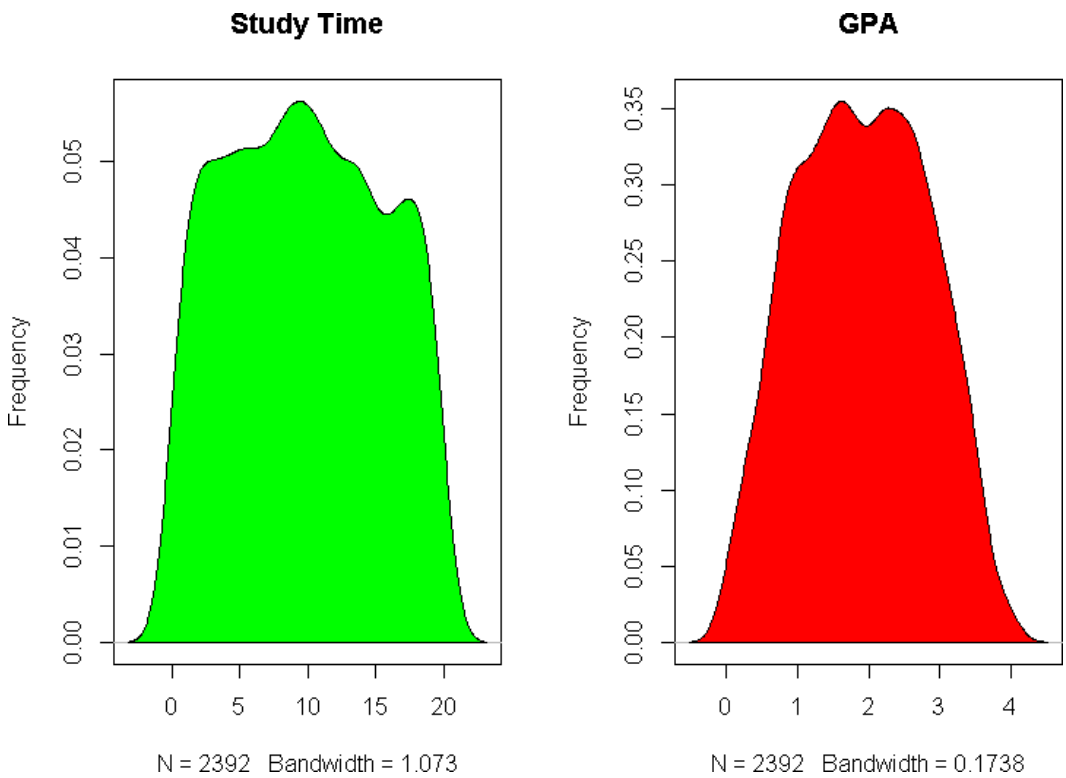

Figure 2: Density plot of variables

Study Time Distribution: The density plot for study time reveals a slightly negatively skewed distribution, indicating that most students study between 5 to 15 hours per week, with fewer students dedicating very high or very low hours to studying.

GPA Distribution: The density plot for GPA shows a moderately negatively skewed distribution, with the majority of students clustered around a GPA of 1.89. This indicates that most students achieve GPAs near the median, with fewer students at the extremes.

```
> par(mfrow = c(1, 2))
>
> boxplot(df$StudyTimeWeekly, main = "Study Time")
> boxplot(df$GPA, main = "GPA")
>
```

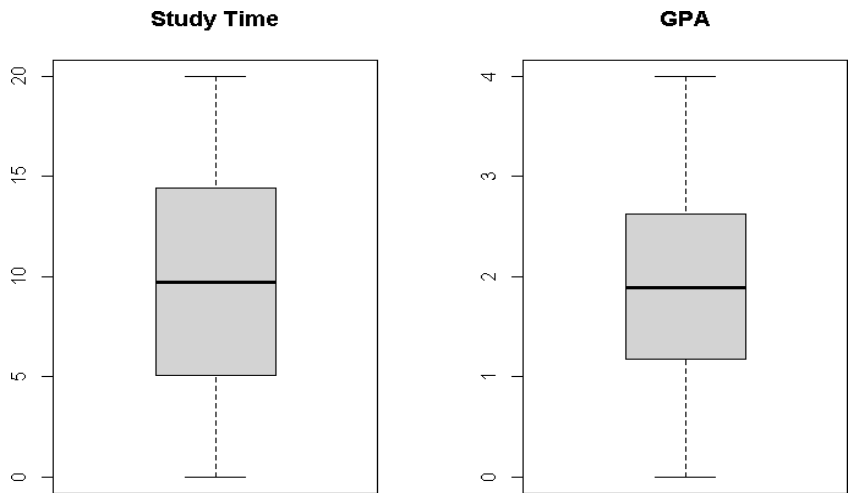

Figure 3: Boxplots of variables

The boxplot for study time demonstrates a well-distributed range between 0 and 20 hours, with no extreme outliers. Most students fall between 5 and 15 hours, representing a balanced study pattern.

The boxplot for GPA shows a consistent range between 0 and 4, with no extreme outliers. Most students achieve GPAs between 1.2 and 2.6, indicating that performance is generally concentrated around these values.

```
> # Summary statistics
> summary(df)

StudyTimeWeekly          GPA
 Min.   : 0.001057   Min.   :0.000
 1st Qu.: 5.043079   1st Qu.:1.175
 Median : 9.705363   Median :1.893
 Mean   : 9.771992   Mean   :1.906
 3rd Qu.:14.408410   3rd Qu.:2.622
 Max.   :19.978094   Max.   :4.000
```

```
> # Correlation
> cor(df)

                StudyTimeWeekly       GPA
StudyTimeWeekly       1.0000000 0.1792751
GPA                   0.1792751 1.0000000
```

To quantify the relationship between study time and GPA, a correlation analysis and a simple linear regression model were performed. The correlation coefficient between study time and GPA is modest (0.179), indicating a weak but positive linear association. The regression model, which uses study time as the predictor for GPA, yields a statistically significant positive coefficient for study time (0.029, $p < 0.001$). This indicates that for every additional hour spent studying weekly, the GPA is predicted to increase by approximately 0.029 points, although this effect is small. The model's R^2 value is 0.032, suggesting that study time explains only a small fraction (3.2%) of the variance in GPA. The remaining 97% of the variability remains unexplained, suggesting that other factors, such as individual differences, learning styles, and external circumstances, may play a more significant role in determining GPA.

```
> # Regression model
> m1 <- lm(GPA ~ StudyTimeWeekly, data = df)
> summary(m1)
```

```
Call:
lm(formula = GPA ~ StudyTimeWeekly, data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.99922 -0.73441 -0.00121  0.72442  2.31142 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)     1.622567   0.036777  44.119   <2e-16 ***
StudyTimeWeekly 0.029024   0.003258   8.909   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.9005 on 2390 degrees of freedom
Multiple R-squared:  0.03214,	Adjusted R-squared:  0.03173 
F-statistic: 79.36 on 1 and 2390 DF,  p-value: < 2.2e-16
```

The residuals from the regression model appear to be approximately normally distributed, as shown in Figure 4, although there is some deviation in the tails. Overall, the analysis provides evidence of a weak positive relationship between study time and GPA. However, the low explanatory power of the model suggests that further research should consider additional factors that may contribute to academic success. Additionally, Figure 4 shows a random scatter of points around the horizontal axis, suggesting that the model's assumptions are met and that the residuals are homoscedastic (i.e., have constant variance). Finally, The RMSE represents the average magnitude of the prediction errors, with lower values indicating better model performance. In this case, an RMSE of approximately 0.91 means that, on average, the model's predictions are off by about 0.91 GPA points from the actual values. This is a relatively poor result, especially when the GPA values are within the range 0-4.

```
> # Model diagnostics
>
> par(mfrow = c(1, 2))
> plot(m1, which = 2)
> plot(m1, which = 1)
```

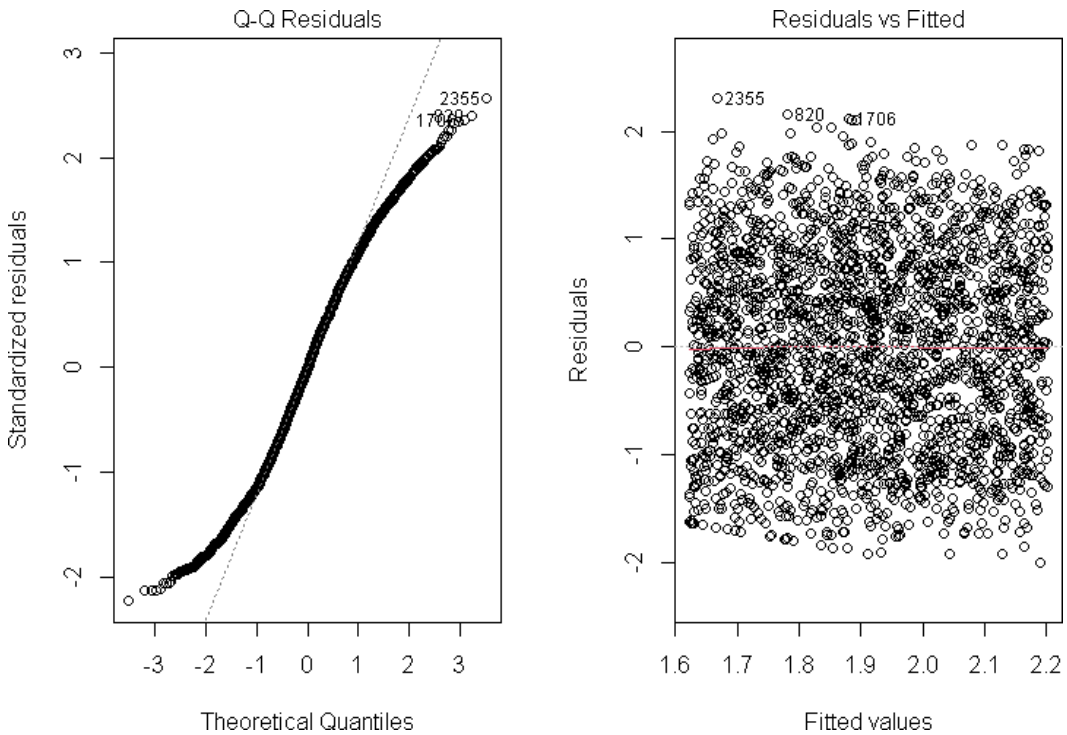

Figure 4: Model assumption checks

Using the model, predicted GPAs were calculated for students studying 3, 5, 12, 15, and 19 hours weekly. Predicted values ranged from 1.71 (3 hours) to 2.17 (19 hours), illustrating the model's modest sensitivity to changes in study time. The RMSE (Root Mean Squared Error) was computed as 0.9001, indicating that the model's average prediction error is around 0.9 GPA units. This error is substantial relative to the range of GPA values (0 to 4), highlighting the model's limited practical utility. The analysis confirmed a statistically significant but weak relationship between weekly study time and GPA. While the results suggest that increasing study time can improve GPA, the low R^2 and high RMSE highlight that other unmeasured factors likely play a more substantial role in determining academic performance. Future work could explore additional variables or employ more complex models to better capture the factors influencing GPA. Therefore, while studying more hours may slightly improve GPA, it is not a strong predictor of academic success.

```
> # Make some predictions
> study_hours = data.frame(StudyTimeWeekly = c(12, 15, 3, 5, 19))
> study_hours$pred_gpa = predict(m1, study_hours)
> print(study_hours)

  StudyTimeWeekly pred_gpa
1              12 1.970851
2              15 2.057923
3               3 1.709638
4               5 1.767685
5              19 2.174018
```

```
> # Check model performance
> Actual = df$GPA
> Predicted = predict ( m1 , newdata = df)
>
> # Calculate RMSE
> RMSE =  sqrt ( sum ((Actual - Predicted) ^2, na.rm = T) /nrow(df) )
> paste0("The RMSE is ", RMSE)

[1] "The RMSE is 0.900141145032665"
```

Conclusion:

Based on the statistical analysis, the results indicate a weak positive relationship between weekly study hours and GPA. While study hours have a slight impact on improving GPA, this effect is very limited, as the model explains only 3.2% of the variance in GPA. These findings suggest that other factors, such as study quality, learning environment, and personal motivation, play a much larger role in academic performance.

Therefore, it can be concluded that increasing study hours may be beneficial for academic improvement, but it is neither the sole nor the most significant factor. These results highlight the importance of exploring additional variables in future studies to provide a more comprehensive and accurate understanding of academic performance.